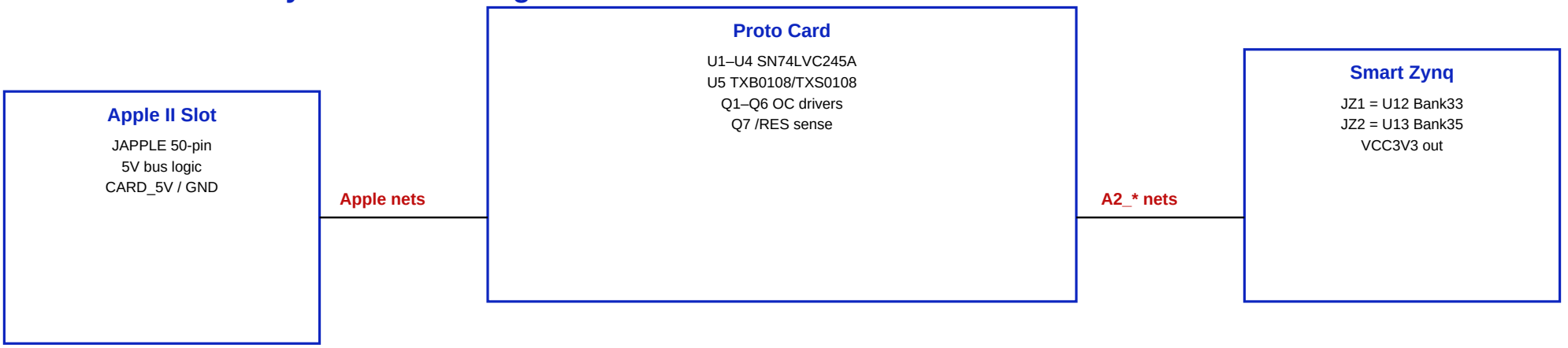
Red Proto Card — System Block Diagram



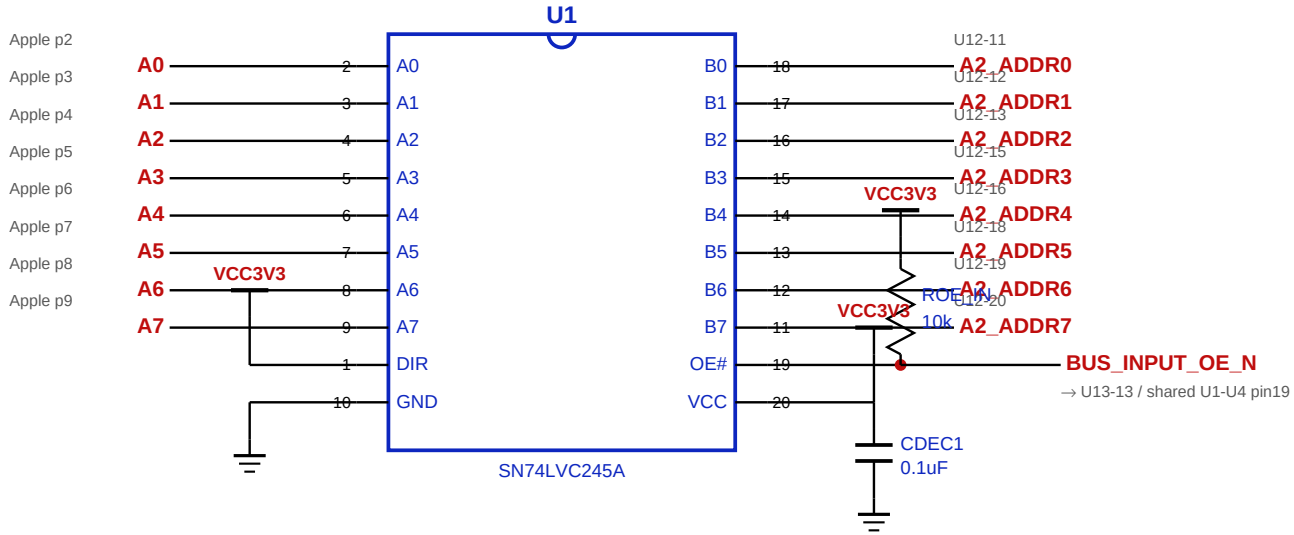
POWER DOMAINS
- CARD_5V = Apple pin 25 → U5 VCCB + 15k/22k sense only
- VCC3V3 = Zynq JZ1/JZ2 pins 37–38 → U1–U4 VCC, U5 VCCA, pull-ups, Q7 pull-up
- GND = Apple pin 26 ↔ JGND (≥4 wires) ↔ Zynq GND
- JZ 39–40 board +5V = DEFAULT NOT CONNECTED

SHEETS
- 2–5 U1–U4 6 U5 TXB 7–8 Q1–Q6 9 Q7+5V sense
- 10 JAPPLE 11 JZ1/U12 12 JZ2/U13 13 Rules / short check

A2N20v2 / Smart Zynq Proto Adapter

A2N20v2 Red Proto Card *from Smart Zynq style*		
Block Diagram		
Size: A4	Rev: 3.0	1/13

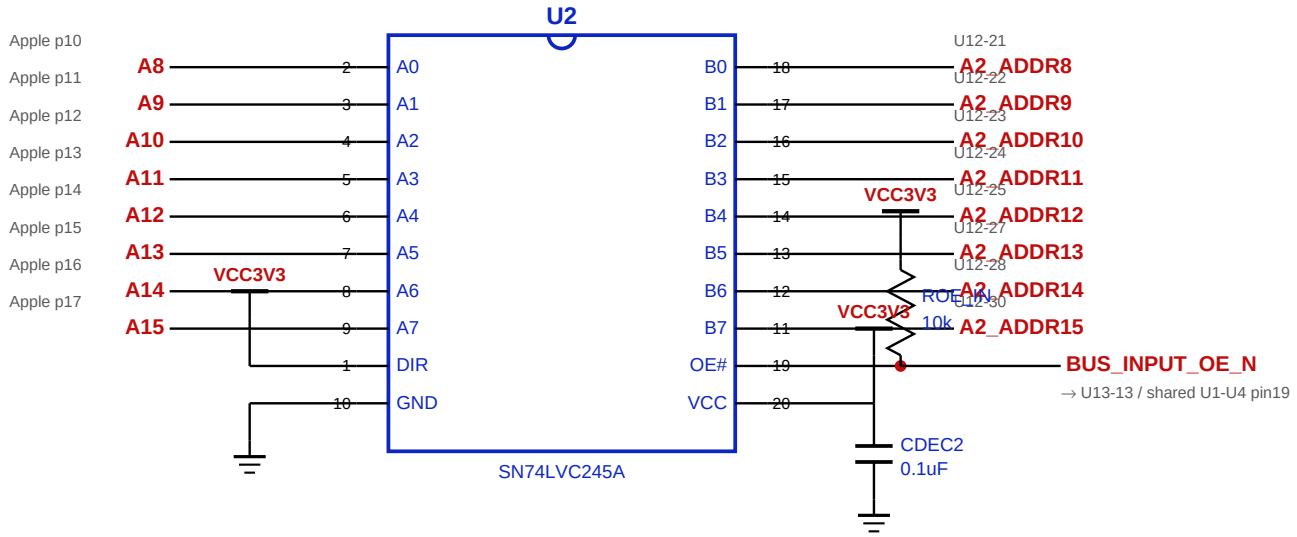
U1 SN74LVC245A



Style matched to Smart_Zynq_SP_schematic.pdf (HelloFPGA / Altium).
Connectivity is by identical RED net names across sheets.
DIR=1 (A→B Apple→Zynq). A-side 5V-tolerant. VCC=+3V3 only — never CARD_5V.

A2N20v2 Red Proto Card *from Smart Zynq style*		
U1 Address A0-A7		
Size: A4	Rev: 3.0	2/13

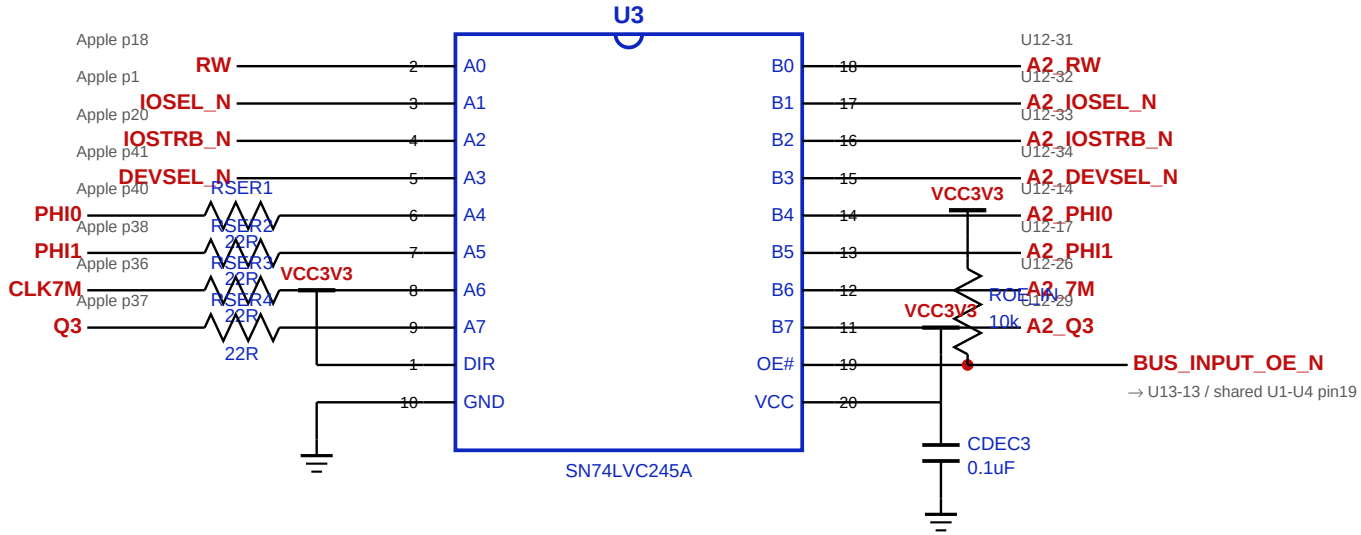
U2 SN74LVC245A



Style matched to Smart_Zynq_SP_schematic.pdf (HelloFPGA / Altium).
Connectivity is by identical RED net names across sheets.
DIR=1 (A→B Apple→Zynq). A-side 5V-tolerant. VCC=+3V3 only — never CARD_5V.

A2N20v2 Red Proto Card *from Smart Zynq style*		
U2 Address A8-A15		
Size: A4	Rev: 3.0	3/13

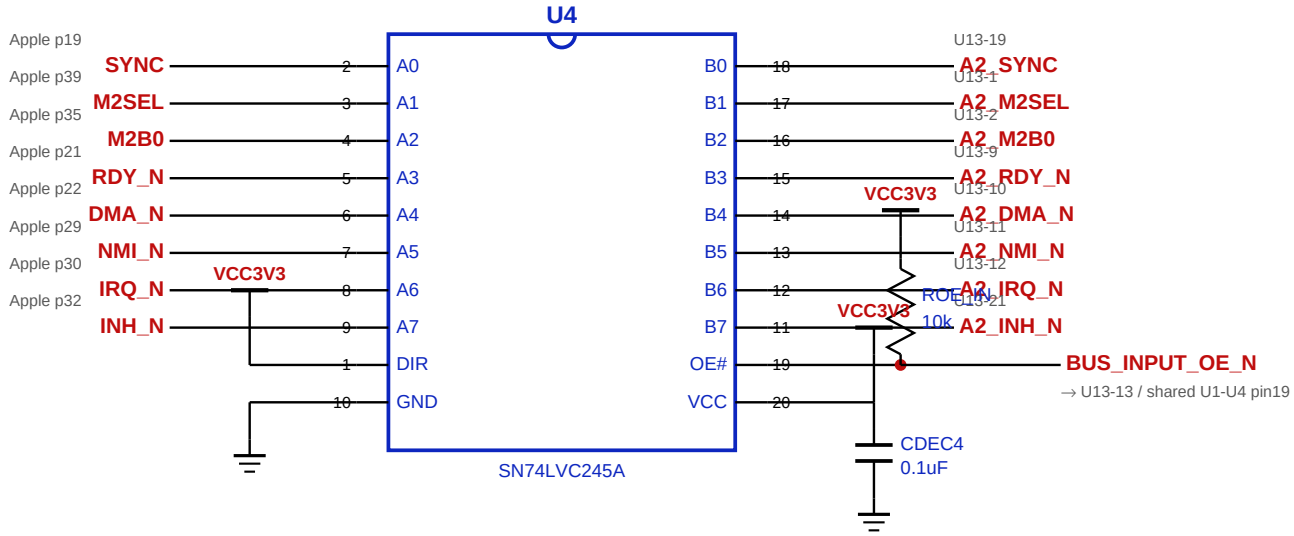
U3 SN74LVC245A



Style matched to Smart_Zynq_SP_schematic.pdf (HelloFPGA / Altium).
Connectivity is by identical RED net names across sheets.
DIR=1 (A→B Apple→Zynq). A-side 5V-tolerant. VCC=+3V3 only — never CARD_5V.

A2N20v2 Red Proto Card *from Smart Zynq style*		
U3 Controls/Clocks		
Size: A4	Rev: 3.0	4/13

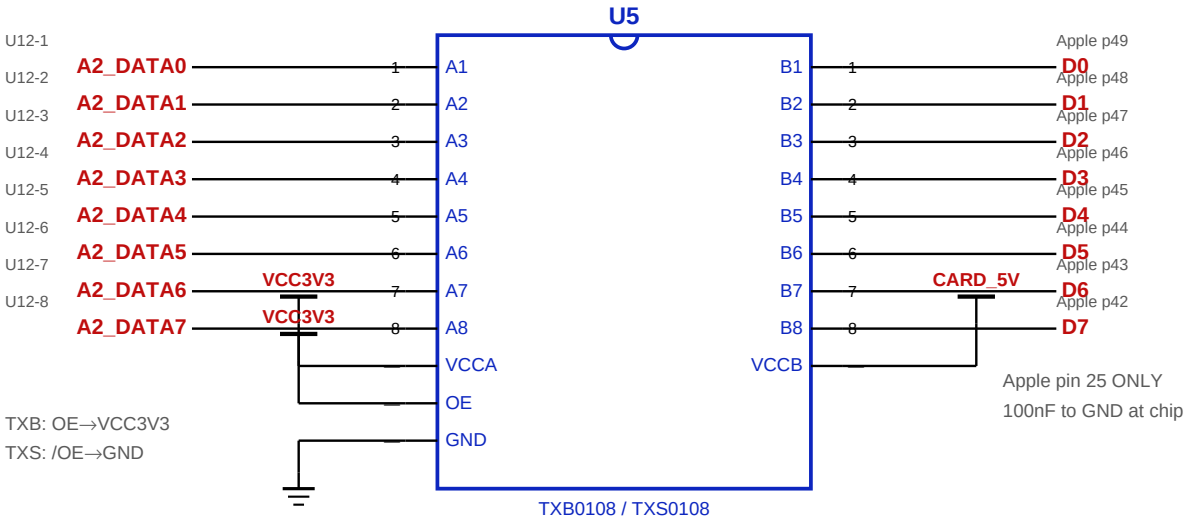
U4 SN74LVC245A



Style matched to Smart_Zynq_SP_schematic.pdf (HelloFPGA / Altium).
Connectivity is by identical RED net names across sheets.
DIR=1 (A→B Apple→Zynq). A-side 5V-tolerant. VCC=+3V3 only — never CARD_5V.

A2N20v2 Red Proto Card *from Smart Zynq style*		
U4 Shared Senses		
Size: A4	Rev: 3.0	5/13

U5 TXB0108 / TXS0108 — D0–D7



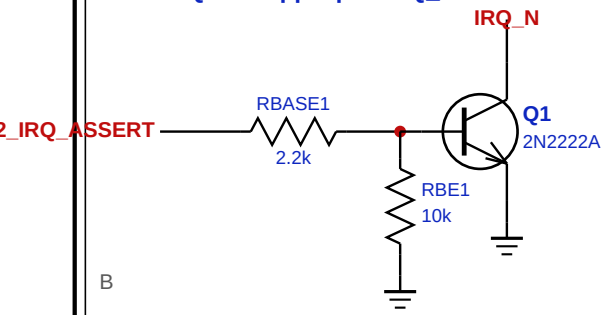
Legacy U12-9 (A2_DATA_DIR) and U12-10 (A2_DATA_OE_N): keep 10k to VCC3V3; do NOT wire to TXB OE.
SHORT RISK: CARD_5V (VCCB) must never continuity-beep to GND.

A2N20v2 Red Proto Card *from Smart Zynq style*		
U5 Data Bus TXB0108		
Size: A4	Rev: 3.0	6/13

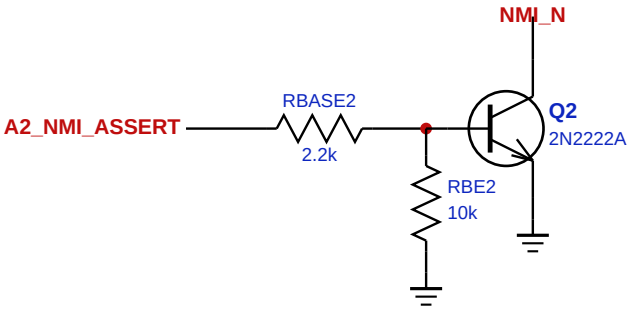
Q1–Q3 Open Collector

Zynq HIGH → NPN ON → Apple line LOW (open-collector). Emitters to GND.

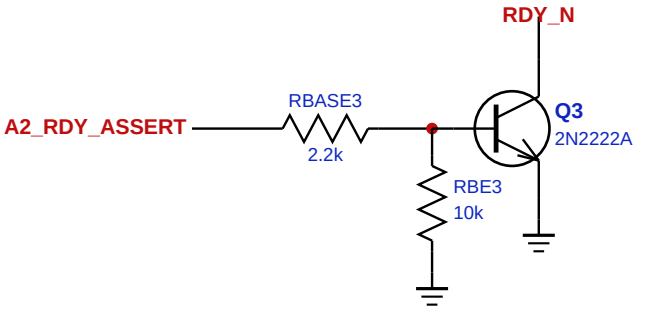
Q1 → Apple p30 IRQ_N



Q2 → Apple p29 NMI_N



Q3 → Apple p21 RDY_N

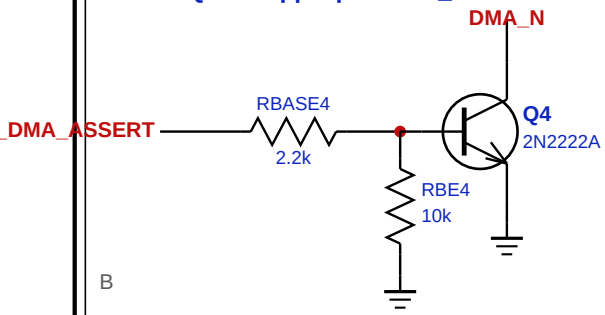


A2N20v2 Red Proto Card *from Smart Zynq style*		
Q1–Q3 Open Collector		
Size: A4	Rev: 3.0	7/13

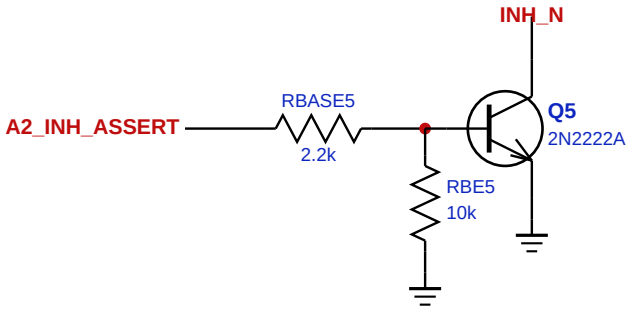
Q4–Q6 Open Collector

Zynq HIGH → NPN ON → Apple line LOW (open-collector). Emitters to GND.

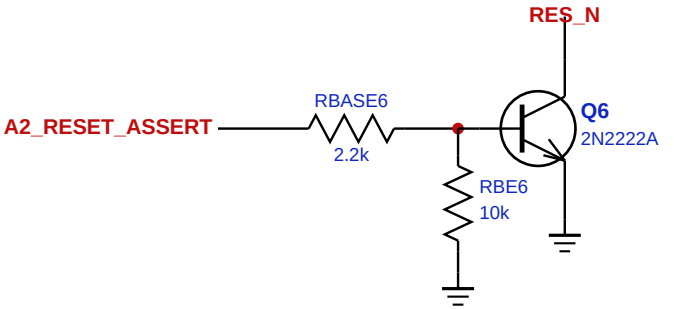
Q4 → Apple p22 DMA_N



Q5 → Apple p32 INH_N

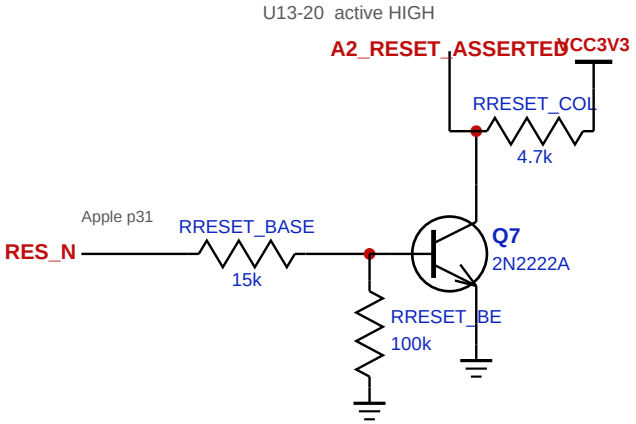


Q6 → Apple p31 RES_N

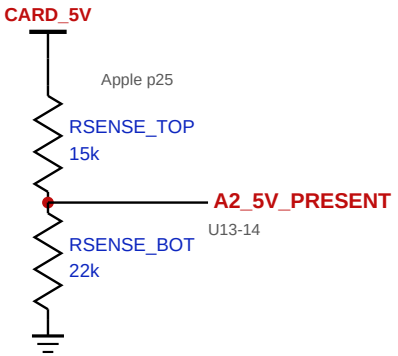


A2N20v2 Red Proto Card *from Smart Zynq style*		
Q4–Q6 Open Collector		
Size: A4	Rev: 3.0	8/13

Q7 /RES sense (inverting)



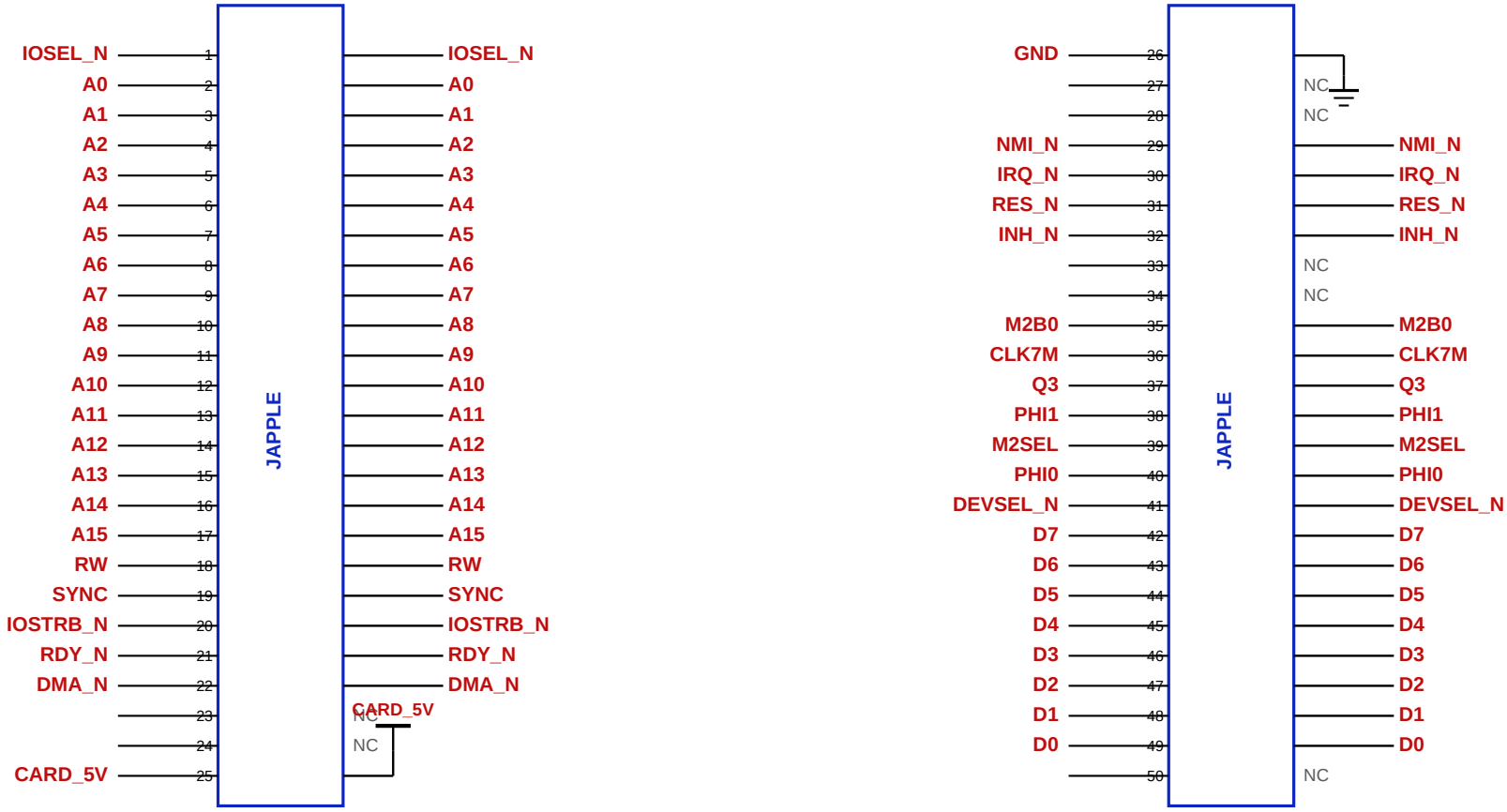
Apple +5V presence sense



RES_N high → Q7 ON → A2_RESET_ASSERTED low. RES_N low → Q7 OFF → pulled high by 4.7k.
Q6 collector also on RES_N (open-collector share is intentional).
Sense divider alone ≈ 37kΩ. Continuity beep CARD_5V↔GND = HARD SHORT — do not power Apple.

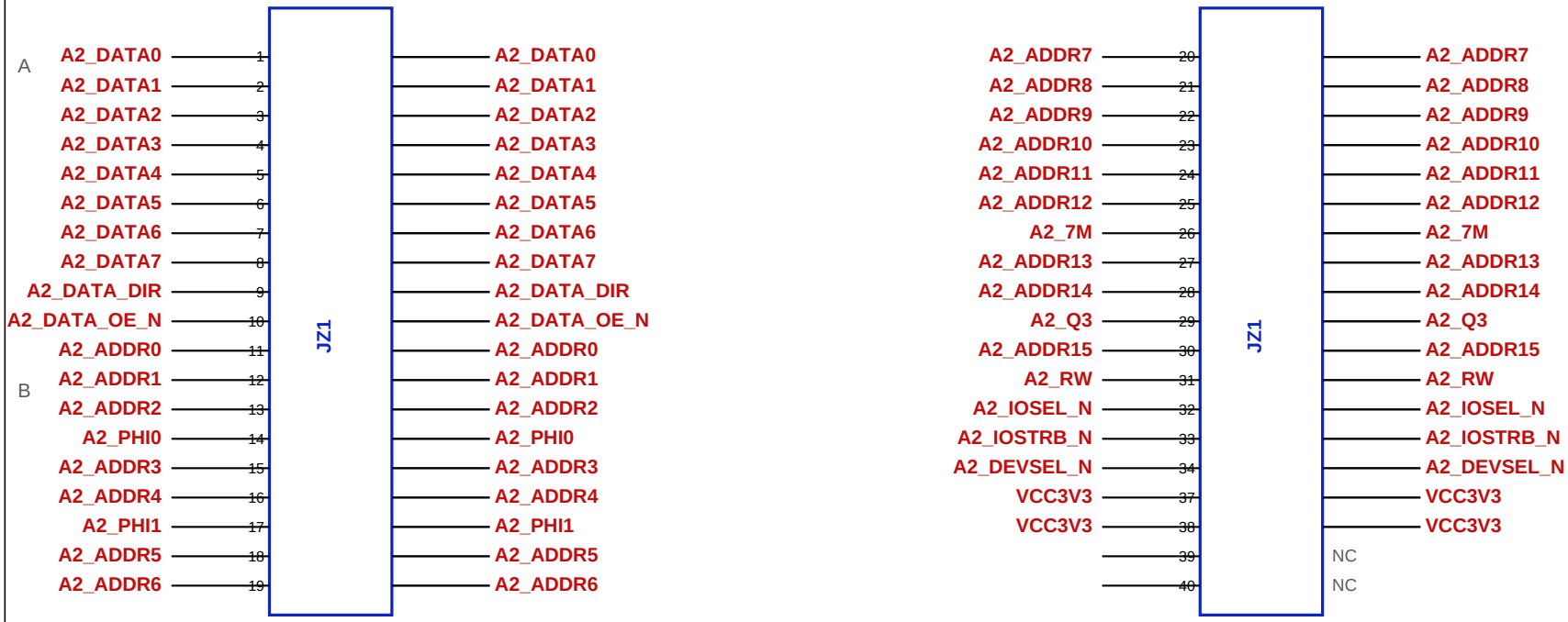
A2N20v2 Red Proto Card *from Smart Zynq style*		
Q7 Reset Sense + 5V Presence		
Size: A4	Rev: 3.0	9/13

JAPPLE Apple II Edge



A2N20v2 Red Proto Card *from Smart Zynq style*		
JAPPLE		
Size: A4	Rev: 3.0	10/13

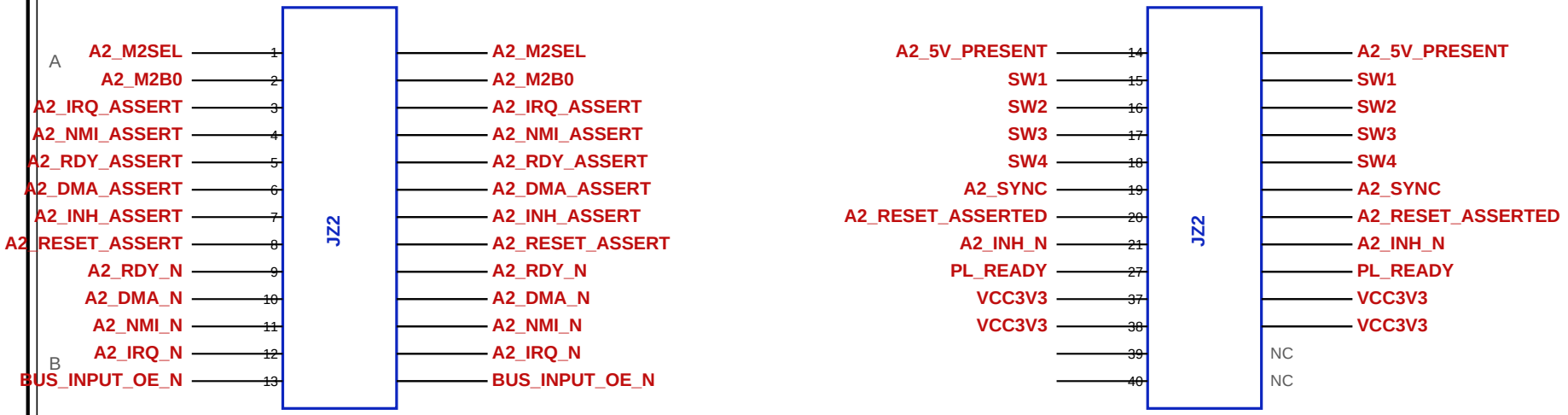
JZ1 U12/J6



P 1	A2_DATA0	U5-A1
P 2	A2_DATA1	U5-A2
P 3	A2_DATA2	U5-A3
P 4	A2_DATA3	U5-A4
P 5	A2_DATA4	U5-A5
P 6	A2_DATA5	U5-A6
P 7	A2_DATA6	U5-A7
P 8	A2_DATA7	U5-A8

A2N20v2 Red Proto Card *from Smart Zynq style*		
JZ1 U12		
Size: A4	Rev: 3.0	11/13

JZ2 U13/J5



P 1	A2_M2SEL	U4-B1
P 2	A2_M2B0	U4-B2
P 3	A2_IRQ_ASSERT	Q1
P 4	A2_NMI_ASSERT	Q2
P 5	A2_RDY_ASSERT	Q3
P 6	A2_DMA_ASSERT	Q4
P 7	A2_INH_ASSERT	Q5
P 8	A2_RESET_ASSERT	Q6

A2N20v2 Red Proto Card *from Smart Zynq style*

JZ2 U13		
Size: A4	Rev: 3.0	12/13

Assembly rules & 5V short checklist

REQUIRED PARTS (manual BOM — often missed on first build)
ROE_IN 10k on BUS_INPUT_OE_N; ROE_DATA 10k on U12-10; RDIR_DATA 10k on U12-9
RSENSE_BOT 22k; CDEC 0.1uF on each LVC245; 0.1uF on TXB VCCA and VCCB

NC APPLE PINS: 23, 24, 27, 28, 33, 34, 50
JZ1/JZ2 pins 39–40 (+5V board) DEFAULT NC

TXB0108 OE = active HIGH → VCC3V3 TXS0108 /OE = active LOW → GND

5V ↔ GND SHORT (power OFF)
1. Card alone, ribbons off: meter Apple finger 25 to 26 — must NOT beep
2. If beep: bridge 25/26, TXB VCCB wiring, or dead U5 — lift VCCB and retest
3. If open alone but shorts with ribbons: disconnect JZ 39/40 and any 5V feed
4. Sense path alone is ~37kΩ (never a continuity beep)

Netlist sources: hardware/*.csv + docs/A2N20v2_Smart_Zynq_FINAL_Manual.pdf
Visual style reference: references/Smart_Zynq_SP_schematic.pdf

A2N20v2 Red Proto Card *from Smart Zynq style*		
Rules / Short Checklist		
Size: A4	Rev: 3.0	13/13